Module 2: Frequency Tables and Charts — Practice

Lesson 2 | Construct, check, interpret

Table of contents

Preparation 1
1. Reconstruct the soft-drink summary 1
2. Choose and defend a categorical chart 2
3. Group audit durations 2
4. Investigate grouping choices 2
5. Read the distribution 2
6. Write for a manager 3

Preparation

Read Lesson 2. Use the CSV bundle and data guide. Open the 20261 course project when using the paths below; adjust paths if working from a separate folder.

Submit a short Quarto report with reproducible R code, labeled tables and figures, and interpretations. Use the course style. Answer questions about the observed data before making recommendations.

```
# library(): loads a package so its functions can be used in this session.
# read.csv() imports a CSV into a data frame with named columns.
# The relative path starts at the course project folder; the source CSV is not modified.
library(ggplot2)
soft <- read.csv("data/module02/soft_drinks.csv")
audit <- read.csv("data/module02/audit_times.csv")
```

1. Reconstruct the soft-drink summary

Use `brand_book_label`, not the two brand columns stacked together.

1. State the observational unit and measurement level of the brand variable.
2. Make a table of counts, relative frequencies, and percentages for all five brands.
3. Check that the counts and proportions have the expected totals.
4. Identify the largest share, the tied categories, and the combined share of the three most frequent brands.
5. Explain why the two brand columns do not represent 100 separate purchases.

2. Choose and defend a categorical chart

1. Draw a horizontal bar chart ordered by descending frequency. Include readable labels and a zero baseline.
2. Draw a flat pie chart of the same percentages.
3. Explain which chart you would give a manager who wants to compare the two least frequent brands, and why.
4. Explain why the statement “Coca-Cola has 38% of the national market” is not supported by this dataset.
5. Suppose another sample contains 200 purchases. Explain why comparing raw counts alone could mislead; you do not need to invent or analyze a second dataset.

3. Group audit durations

1. Define five nonoverlapping intervals of width five days, starting at 10 and ending at 35. State which endpoints are included.
2. Make frequency and percentage tables using these intervals.
3. Draw a histogram with the same boundaries. Verify that its bar counts agree with your table.
4. Identify the modal class. Explain why it does not necessarily identify the most frequent exact duration.
5. Make a cumulative table. Report the count and percentage taking less than 25 days, and the count taking at least 25 days.
6. Use the raw CSV to compare the counts for **less than 20 days** and **20 days or less**. Explain the difference.

4. Investigate grouping choices

Use wider intervals **[10,20), [20,30), [30,40)**.

1. Recompute the frequencies and draw a second histogram.
2. Confirm that both histograms contain the same 20 audits.
3. Describe one detail that is less visible with wider bins.
4. Explain why a new set of bins does not mean that the underlying observations changed.
5. A colleague proposes unequal-width bins while keeping counts as heights. Explain why the area of each rectangle matters and what quantity should determine its height.

5. Read the distribution

1. For the Audit histogram, write one sentence for each feature: **shape, center, spread, and unusual features**. Distinguish approximate visual readings from exact CSV calculations.
2. Explain how you decide whether a histogram is right-skewed or left-skewed. Does the direction come from its tallest bar or its longer tail?
3. Compare the artificial unimodal, bimodal, and multimodal illustrations in the lesson. What might multiple peaks suggest? Give two reasons they do not prove that distinct populations exist.
4. Explain why a roughly symmetric histogram does not establish normality and why an isolated bar does not establish a data error.
5. Compare the 1-, 5-, and 10-day Audit histograms. Which features become harder to see in wider bins? Does the 33-day audit disappear from the dataset?
6. Optional: use the student age/height CSV to reproduce the comparison. Describe the age tail and the height concentration. Read the source notes and explain why the charts retain recorded units and do not support claims about all university students.

6. Write for a manager

Write two short paragraphs, each about 60–90 words:

- **Purchases:** summarize the leading brands and recommend a useful next information-gathering step before a purchasing decision. Distinguish an observed purchase share from profit or national demand.
- **Audit durations:** summarize the modal class and the share below 25 days. State what these observations do and do not justify when planning future work.

Do not claim that a descriptive chart establishes causation, guarantees completion times, or proves that an organization is inefficient.

Return to Lesson 2.